

Seminar 1 Getting to know you

Industry Applications of Artificial Intelligence Systems

Bauman Moscow State Technical University



Medicine



Finance



Transport



Manufacturing



Development / IT



Logistics



Science



Cybersecurity



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AI enthusiast and researcher • Builder of AI-based systems

I build innovative solutions, automate processes, and deploy leading-edge AI technologies to solve real business and user problems.



Clarifian clarifian.io

AI-based automatic diagram drawing. Turn text descriptions and ideas into structured, clear visual diagrams and charts in one click.



AI Mentor mentor.tellian.ru

AI-based testing and mentoring. Personalized learning, skills assessment, and intelligent support along your development path.



w4check.io

Reliability checks for crypto counterparties.



workain.ai

AI automation of workflows and tasks.



Blog · tellian.io

Original writing on technology, AI, and software development.

17 lectures × 3 modules — from fundamentals to AI by industry

Today



Module 1

Fundamentals and familiar industries

Lectures 1–6

Module 2

High-tech manufacturing

Lectures 7–12

Module 3

ICT, science, extraction, synthesis

Lectures 13–17

MT1 · S6

MT2 · S12

MT3 · S17

Cross-cutting themes — in every lecture and seminar



Choosing a tool

not every AI fits every task



Common mistakes

and how to avoid them



Human / AI accountability

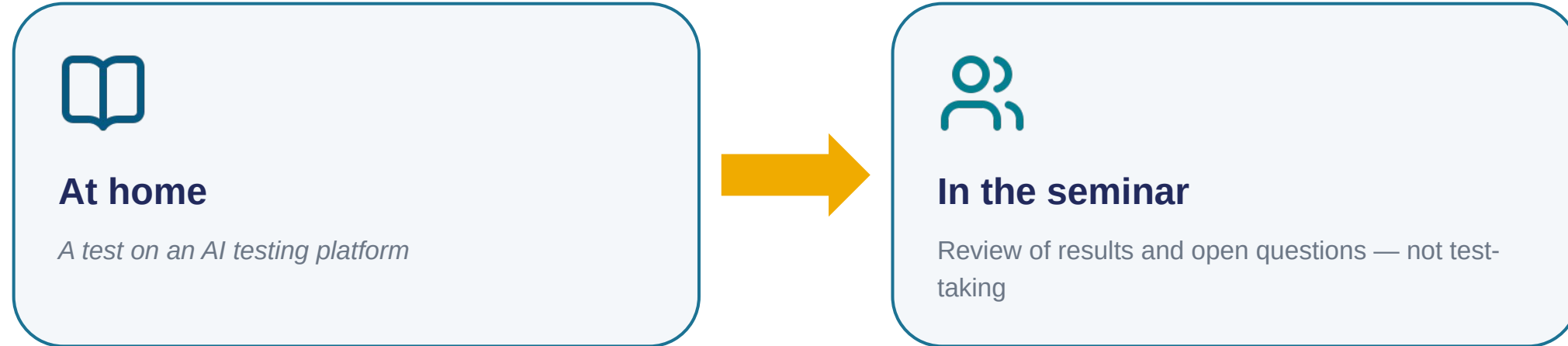
who's responsible for the result



Info security

what can and can't be shared with AI

Midterms — test at home, review in the seminar



Midterm · Seminar 6

Midterm · Seminar 12

Midterm · Seminar 17

Midterm seminar: not test-taking — reviewing what raised questions

MAX course chat — everything important in one place

Group chat: “Industrial Use of AI 2026”



Scan with your phone's camera

[max.ru/join/
sHoHlHl4jvW_swcq5ZLowvz6G94m0IMhAwngxuxsEpl](https://max.ru/join/sHoHlHl4jvW_swcq5ZLowvz6G94m0IMhAwngxuxsEpl)



Announcements

Schedule, changes, and important course news — all in one place.



Lecture and seminar materials

Slides and materials appear in the chat as sessions happen.



Questions between sessions

You can ask a course question and get an answer without waiting for class.

Warm-up: how do you use AI?

We answer out loud and discuss — no camera, no counting. We'll compare your answers with outside statistics in the next block.



Which AI tool do you use most often?

ChatGPT · GigaChat/YandexGPT · Copilot/Claude/DeepSeek · none



How often do you use AI?

every day → weekly → monthly → less often → never



How much do you trust AI's accuracy?

a scale from 1 to 5



Has AI ever clearly let you down?

yes · no · not sure

Used more, trusted less — and that's not a paradox

USAGE

84%

use or plan to use AI tools

2024 → 2025: 76% → 84% (+8 pp)

WHAT THEY USE MOST

ChatGPT 82%

GitHub Copilot 68%

Gemini 47%

Claude 41%

% among those who use AI

TRUST

29%

trust the accuracy of what AI produces

was 40% in 2024 (–11 pp)

46% explicitly don't trust it · only 3% trust it fully

Russia · VCIOM, 2026

78% used neural networks over the past year (was 73%) · 58% — at least once a week (was 51%)

64% believe AI is beneficial, 67% of them — only in certain areas · N=3209, 18+, 25–27.06.2026, ≤±1.7%

Who has a story — did it help or let you down?



Hallucinated fact



Outdated data



A task solved only
on the surface — looks right,
dig in — it's not



Lost context in a conversation



Mismatched tone



Answer doesn't
fit your situation

Who wrote it — fiction: AI or human?

Text A

“How strange — he died, and we live on... Except I suspect that every time we go to sleep, we die in exactly the same way. And the sun goes away forever, and the whole of history ends. And then non-being gets bored with itself, and we wake up. And the world arises again.”

Text B

“Sleep is a rehearsal for oblivion that we go through every night without noticing the exam. We turn off the light and, for a few hours, stop existing for ourselves, and in the morning we agree that it was an intermission, not a cancellation. Maybe the only difference between sleep and death is that after sleep, someone is waiting for us again.”

Text A — Pelevin, “Generation P”. Text B — AI



Victor Pelevin, “Generation P” (1999)

“How strange — he died, and we live on... Except I suspect that every time we go to sleep, we die in exactly the same way. And the sun goes away forever, and the whole of history ends...”



Generated by AI, prompted for a similar style

“Sleep is a rehearsal for oblivion that we go through every night without noticing the exam. We turn off the light and, for a few hours, stop existing for ourselves...”



Style can be copied, but a short fragment doesn't always reveal the author



A long work: plot, composition, the development of an idea — harder to imitate



Short fragment: AI imitates sentence rhythm and tone more convincingly

Who wrote it — technical text: AI or human?

Text A

“The while construct in Python runs the loop body for as long as the condition stays true — unlike for, there's no built-in iteration over a sequence here: the interpreter simply checks the condition before every iteration. This is useful when the number of repetitions isn't known in advance and depends on the outcome of computations inside the loop itself.”

Text B

“The for statement is used to iterate over the items of a sequence — for example, a string, tuple, or list — or another iterable object. The list of expressions is evaluated once and must return an iterable object; the resulting set of statements is then executed once for each item, in the order returned by the iterator.”

Text A — AI, in seconds. Text B — official Python documentation



AI, prompted to explain while in a similar style

“The while construct runs the loop body for as long as the condition stays true — unlike for, there's no built-in iteration over a sequence here.”



Python 3 documentation translations (digitology.tech)

“The for statement is used to iterate over the items of a sequence — for example, a string, tuple, or list — or another iterable object. The list of expressions is evaluated once and must return an iterable object; the resulting set of statements is then executed once for each item, in the order returned by the iterator.”

AI version



Faster and often just as accurate for standard language constructs · saves search time

Official documentation



Written and maintained by a team · stable for years

for technical documentation, what matters isn't style or who wrote it, but the accuracy and speed of the answer — AI is especially strong here on typical questions

Who wrote it — code: AI or human?

Code A

```
def prepare_method(self, method):  
    """Prepares the given HTTP method."""  
    self.method = method  
    if self.method is not None:  
        self.method = to_native_string(  
            self.method.upper())
```

Code B

```
def normalize_header_name(name):  
    """Normalize an HTTP header name."""  
    if not name:  
        return name  
    return '-'.join(  
        part.capitalize() for part in  
        name.split('-'))
```

Code A — the requests library, 10+ years in production. Code B — AI, 10-20 seconds

requests, github.com/psf/requests, MIT license

```
def prepare_method(self, method):
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```

AI, prompted to write a similar function

```
def normalize_header_name(name):
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```

requests

Used in millions of projects · battle-tested under real load for 10+ years

AI version

A great fit for typical helper functions in seconds

typical tasks — AI; complex and unique solutions that the system's stability depends on — a human

Who made it — a painting: AI or human?

Image A



Image B



Image B — José María Velasco, 1875. AI would generate something similar in seconds



José María Velasco,
“Valley of Mexico from the Santa Isabel Mountain,” 1875



public domain · Wikimedia Commons

Who made it — a data visualization: AI or human?

Chart A

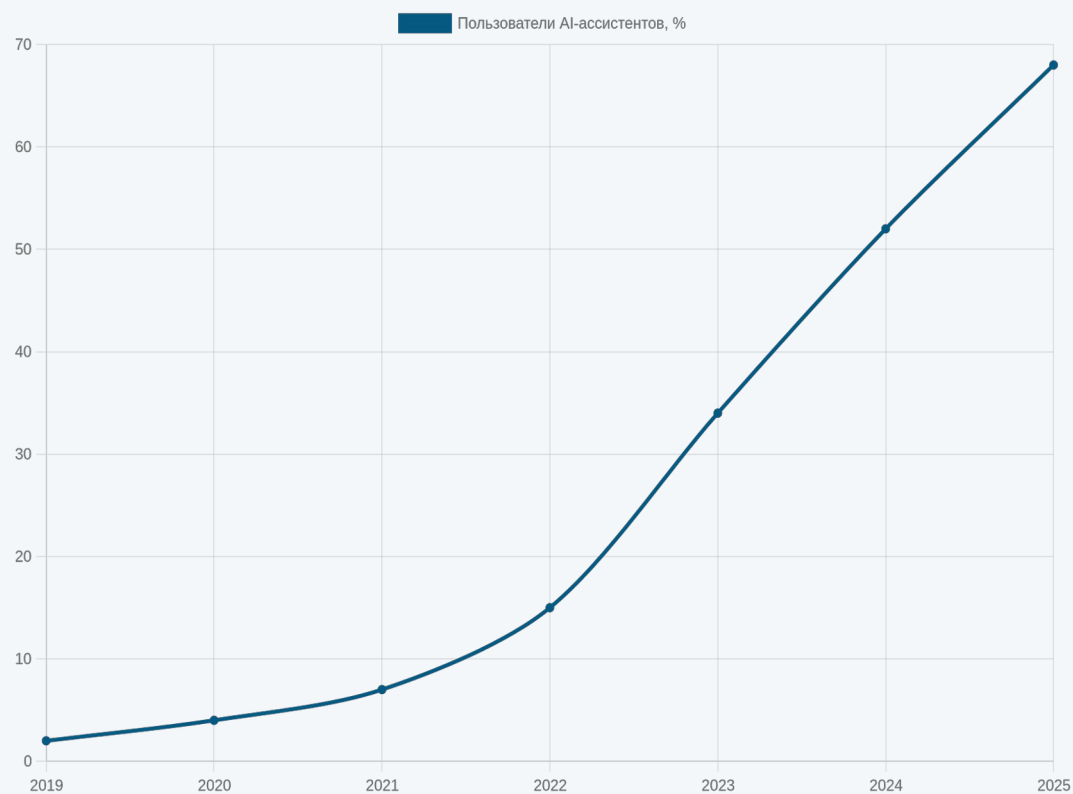


Chart B

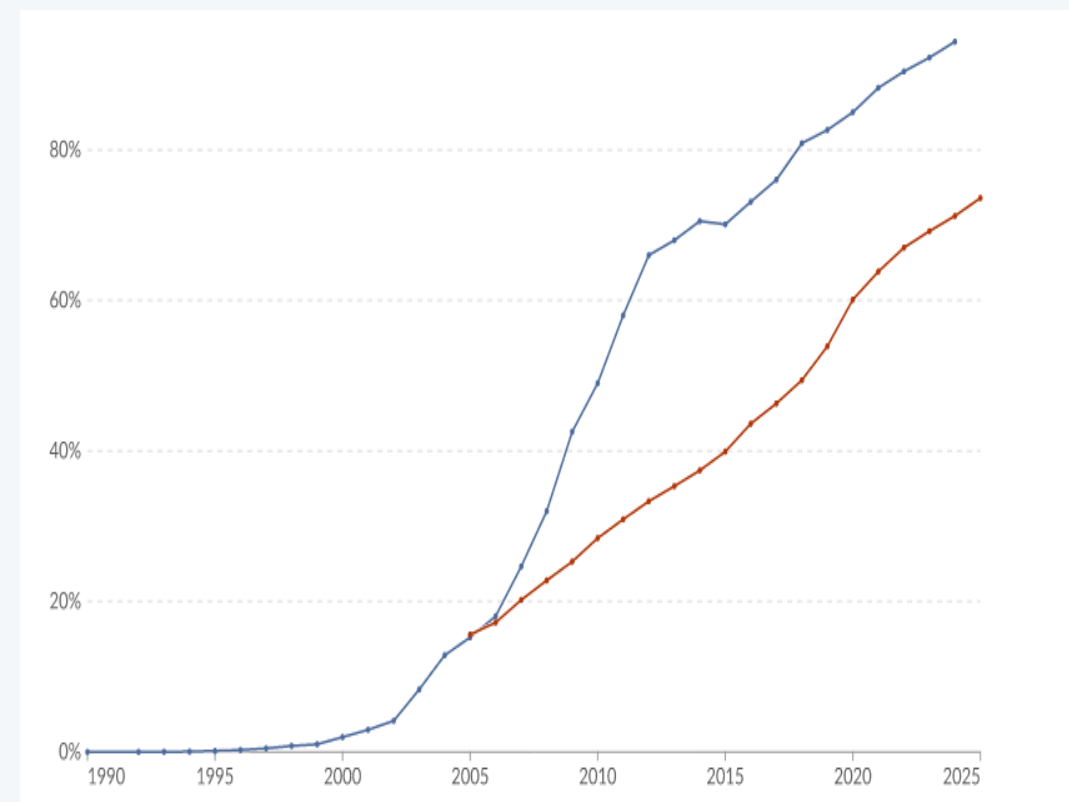
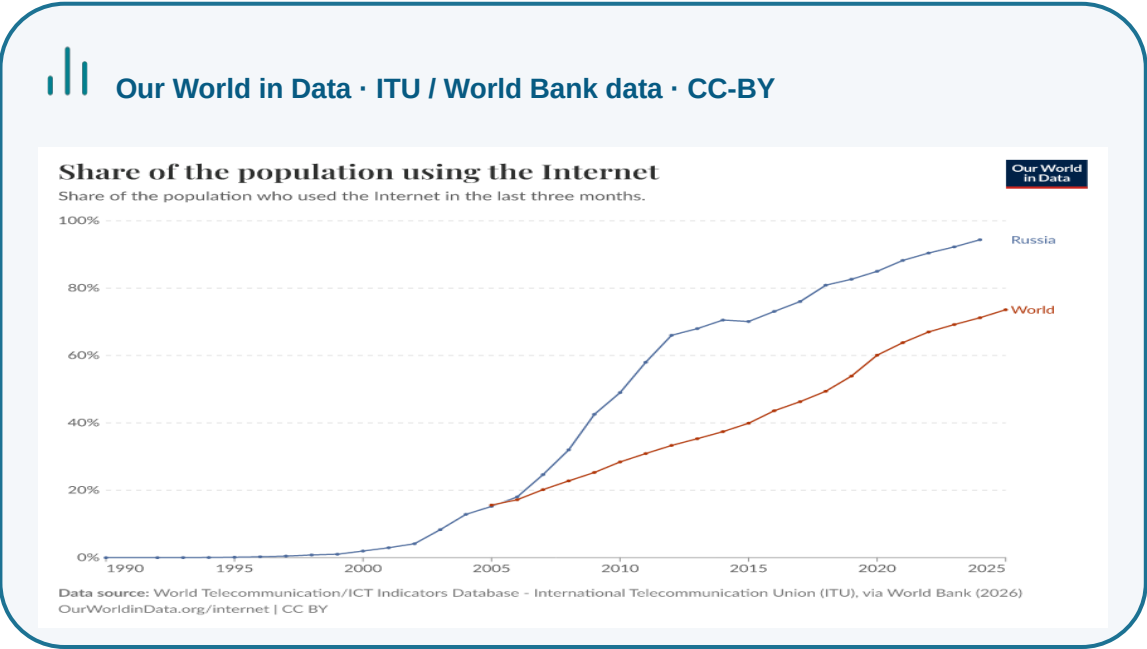
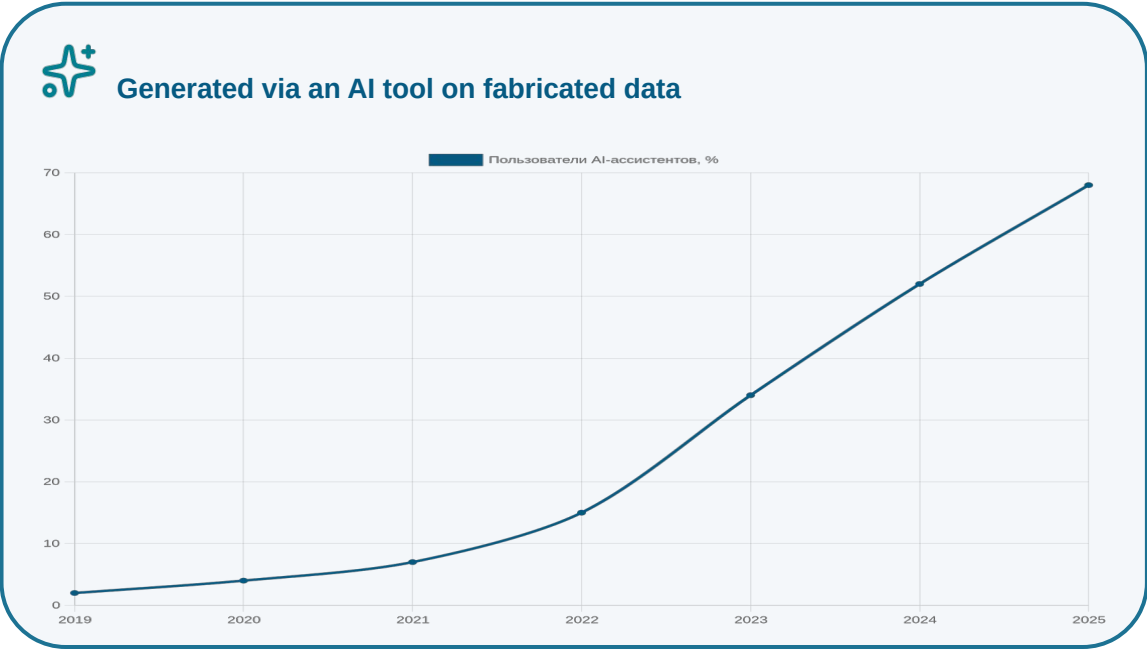


Chart A — AI, seconds. Chart B — Our World in Data, weeks of verification



AI chart

Solves the visualization task in seconds · no question of authorship or ethics here, A research team collects and verifies sources over weeks unlike with the painting

Our World in Data

True or false? Let's check your intuition about AI

-  AI counts the letters in a word directly, the way a human does true | false
-  AI always knows what happened in the world today true | false
-  AI can confidently state something false without meaning to deceive true | false
-  An AI chat can see your personal email and files by default true | false
-  AI always gives an answer consistent with the data it was fed true | false
-  All AI models give the same answer to the same question true | false

vote before the explanation

True or false? Let's check your intuition about AI

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True or false? Let's check your intuition about AI

1  AI counts the letters in a word directly, the way a human does true | ✓ false

2  AI always knows what happened in the world today true | ✓ false

3  AI can confidently state something false without meaning to deceive ✓ true | false

4  An AI chat can see your personal email and files by default true | ✓ false

5  AI always gives an answer consistent with the data it was fed true | ✓ false

6  All AI models give the same answer to the same question true | ✓ false

The “AI reality-check” memo — 5 points to carry into the next week

1

A confident tone in an AI's answer doesn't mean the answer is factually correct

2

AI doesn't know about events after its training cutoff date without web search

3

An AI's answer isn't derived unambiguously from its training data — because generation is random, the same question can get different answers

4

AI sees text as tokens, not letters — which is why it sometimes trips up on letter-by-letter tasks like counting letters in a word

5

Different AI models can give different answers — cross-checking with several sources is sensible



Medicine



Finance



Transport



Manufacturing



Development / IT



Logistics



Science



Cybersecurity

THANK YOU

From today's snapshot to a
systematic look at AI by industry